

Puspita Kaban

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EDUCATION

MS in Environment & Sustainability | University of Michigan, School for Environment & Sustainability, Ann Arbor (MI) | Aug 2025-May 2027 (expected)

- Geospatial data science specialization, GPA 4.00/4.00
- Coursework : Environmental Racism, Health, and Pathway to Justice; Principles of GIS (ArcGIS); Advanced Geovisualization; Remote Sensing; Intermediate Python for Geospatial Data Science; Analysis and Modeling of Environmental Data; Ecology; Industrial Ecology
- Activity : Indonesian Student Association, Rackham Student AI Working Groups, Healthy Cities certification
- Master's capstone project — Participatory GIS Tools for Seagrass Protection and Restoration
 - Client: The Nature Conservancy
 - Co-designed a participatory GIS platform with TNC's Indonesia team to document how coastal communities use and value seagrass ecosystem services. The workflow spans community-based data collection, spatial analysis, and a web map that returns results to local stakeholders in a usable form. Built with Leaflet, R, and ArcGIS Pro.

MITx MicroMasters in Data, Economics, and Development Policy | Massachusetts Institute of Technology (edX), Remote | Sep 2024 – Aug 2025

- Awarded a J-PAL Southeast Asia scholarship for early-career researchers and development practitioners; coursework in data analysis, development policy, microeconomics, and randomized evaluations

BASc in Computational Statistics | Polytechnic of Statistics (STIS), Jakarta (Indonesia) | Aug 2015-Aug 2019

- Thesis : *Optimization of Cheng & Church Biclustering Method with Possibilistic Biclustering Algorithm*
- Coursework : GIS (QGIS), Data Mining & Knowledge Management, Computational Statistics, Data Simulation, Regression Analysis, Time Series Analysis, Multivariate Analysis, Database Management

TECHNICAL SKILLS

- **Geospatial Analysis:** ArcGIS, QGIS, Google Earth Engine, Leaflet, remote sensing, spatial analysis, cartographic visualization, small area estimation
- **Statistical Analysis:** Advanced proficiency in Python (pandas, NumPy, scikit-learn), R (tidyverse), SPSS, SQL, EViews; regression analysis, multivariate analysis, time series, sampling methods, data mining
- **Data Visualization & Reporting:** Tableau, Looker Studio, Adobe Creative Suite (InDesign, Illustrator), Shiny, dashboard development, infographics

EXPERIENCE

Research Assistant | Michigan Institute for Data and AI in Society - University of Michigan, Ann Arbor (MI) | Jan 2026 – present

- Build the first institutional map of AI infrastructure across University of Michigan campuses, integrating and cleaning data from multiple institutional sources
- Develop a publicly accessible, interactive Leaflet map to visualize the university's AI infrastructure

Research Assistant | School for Environment & Sustainability - University of Michigan, Ann Arbor (MI) | Apr 2026 – July 2026

- Mapped land-use impacts of critical-mineral extraction with the **Urban Sustainability Research Group**, part of a global study applying Geo AI — geospatial foundation models and image-segmentation algorithms — to satellite imagery across diverse geographies
- Analyzed 100+ copper mines through data validation and visual inspection to build a reference dataset and verify Geo AI model outputs

Volunteer Monitoring and Evaluation Specialist | Solar Chapter, East Nusa Tenggara (Indonesia) | May 2026 – July 2026

- Audited and redesigned the organization's monitoring-and-evaluation framework across its organization-wide programs, and developed an AI adaptation for the Mengalir.co platform
- Reviewed data quality and data-flow design for the Mengalir.co platform

Research Assistant | LSA Organizational Studies - University of Michigan, Ann Arbor (MI) | Jan 2026 – Apr 2026

- Conducted systematic literature reviews and quantitative data analysis of economic valuation methods for a book project examining their distributional and political consequences

Monitoring, Evaluation & Research Lead | The 1000 Days Fund, Bali (Indonesia) | Jul 2024-Jun 2025

- Developed technical components for household baseline surveys covering 300+ households, including sampling framework, KoboToolbox questionnaire design, data collection protocols, and quality assurance procedures
- Built a semi-automated data-processing pipeline integrating multiple sources in Python, enabling real-time quality assurance and validation checks for program monitoring
- Established results-based management dashboards in Looker Studio tracking key performance indicators (KPIs) across program components, enabling evidence-based decision-making by senior management
- Standardized M&E workflows by developing standard operating procedures (SOPs), data collection protocols, and technical documentation to strengthen institutional knowledge management

Junior Statistician | Statistics Indonesia, North Sumatera (Indonesia) | Dec 2019-Jun 2024

- Managed 5+ data collection operations (25–4,000+ respondents) at district and subdistrict levels in rural Tapanuli Tengah, including budget allocation, recruitment, and field logistics coordination
- Supervised and provided technical backstopping to 50+ field enumerators and supervisors, delivering training on data collection and quality assurance protocols
- Authored 30+ official statistics reports at district and subdistrict levels, presenting socioeconomic indicators and development trends with data visualizations to inform local policy planning